International Islamic University Chittagong
Center for General Education (CGED)
Midterm Examination, Autumn–2025

Course Code: URED-3503 **Course Title: Political Thoughts and Social Behavior**
(For Law faculty: GEED-3101)

Full Marks: 30 ***Time: 1 hour &30 minutes***

Answer all questions. The right side columns contain marks, CLOs, and Bloom's taxonomy domain for each question.

#	Questions	Marks	CLOs	Bloom's taxonomy domain
1	(a) Explain briefly (any four): • Law-making process of the *Majlis-e-Shura.* • Nature of the judiciary before and during the prophetic era. • Common rights and duties of citizens in an Islamic state. • Difference between *Shari'a* and Manmade law. • Organs of government. **Or,** (b) Define Islamic Politics. Explain the basic principles of Islamic political System elaborately.	10	2	Create Remember& Create
2	What is meant by *Shariah*? Analyze the key features of a constitution that follows *Shariah*.	10	2	Remember & Analyze
3	Summarize the qualities of Chief Executive of an Islamic State. Explain the main duties of the Chief Executive.	10	2	Create

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Bismillahir Rahmanir Rahim

International Islamic University Chittagong

Department of Computer Science and Engineering

Mid-Term Examination, Autumn-2025

Course: **CSE-3521 (Computer Architecture)**

Time: **1 hour and 30 Minutes** Marks: **30**

[Answer all ***three*** questions; Figures in the right hand margin indicate Marks]

Q1.	a)	Define Stored-program concept. Draw and describe the computer architecture related to the Stored-program concept. *	1+3	CO1 CO3
	b)	A complex statement contains the six variables **z, a, b, m, n, y**: $$z = (a + b) - (m + n) * y;$$ The variables **z, a, b, m, n and y** are assigned to the registers **\$t0, \$s0, \$s1, \$s2, \$s3 and \$a0** respectively. What is the compiled Assembly code?	3	CO2
	c)	* What is the assembly language statement corresponding to this machine instruction below: 0xba79 ⇒ 16bits ⇒ invalid code ⇒ 32bits	3	CO2
Q2.	a)	Explain Amdahl's Law. A software team at IIUC optimizes a legacy codebase. After profiling, they discover that 50% of the program's execution time can be optimized. They manage to make this part run 2× faster, while the other 50% remains unchanged. * Using Amdahl's Law, what is the overall speedup of the program?	1+4	CO1 CO2
	b)	What is the difference between CISC and RISC computers? By defining the architectures, explain how mobile and desktop chips align with CISC and RISC. Give example of the chips.	3+2	CO1
		Or		
		Explain four design principles for MIPS? Explain any three MIPS instruction formats with fields.	2+3	CO1
Q3.	a)	An old AMD A1 handles 4 requests/second. How many requests will it handle in 5 seconds? By showing the flowchart of the first method of multiplication, describe the steps.	3+2	CO2 CO3
		Or Flowc + Table + Explanation		
		A cloud storage provider charges 12 taka per GB of storage. A company wants to buy 25 GB of storage space. Using the flow diagram of final method of multiplication, calculate the total cost.	3+2	CO2 CO3
	b)	Show the IEEE 754 binary representation of the number $-0.XY_{ten}$ in single and double precision. Indicate how you handled the normalization process. X and Y are your last two digits of student ID. (56)	4+1	CO2

1 → 4
5 → x

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INTERNATIONAL ISLAMIC UNIVERSITY CHITTAGONG

Department of Computer Science & Engineering (CSE)

BSc in CSE, Mid-Term Examination, Autumn 2025

Course Code: **CSE-3527** Course Title: **Compiler**

Full Marks: **30** Time: **1 Hour 30 minutes**

*[Answer **all** the following questions. Some questions may have an option.*
(Figures at right margin illustrate the marks (M), course objectives (COs), and bloom's taxonomy (DL))]

No.	Questions	M	COs	DL

1. a) Define a program with a proper block diagram that is used to compile a program in high-level languages and contrast it with the interpreter. — 2 CO1 U

Or

Explain the working of a language processing system that converts a high-level source program into an executable target code. Illustrate the process with an appropriate system diagram. — 2 CO1 U

b) Consider the following C program: — 8 (4+1+3) CO1 Ap

```
int main(){
 float salary, basic, increment, RATE;
 scanf("%f%f%f", basic, increment, RATE);
 salary = basic + increment * RATE;
 printf("%f\n", salary);
 return 0;
}
```

Translate this C program step by step through the compiler phases. Draw the diagram of the compiler's phases with output translating annotations at each phase and show the appropriate symbol table entries for identifiers, syntax definition in context-free grammar (CFG), and tokenization for each lexeme. [Assume all variables are floating type]

2. a) What do you mean by syntax definition in compiler construction? Describe each of the basic syntax definition components of a language in compiler design. Give syntax definition for the following conditional if-else statements in C programming: — 4 (1+1+2) CO2 Ap

```
if(condtions){
statements;
} else if(condtions){
statements;
}....
else if(condtions){
statements;
}else{
statements;
}
```

Or,

Describe ambiguity in CFG and its impacts on parser design? Consider the following CFG G_1: — 4 (1+1+2) CO2 Ap

$$digit \rightarrow digit + digit \mid digit - digit$$
$$digit \rightarrow digit * digit \mid digit / digit$$
$$digit \rightarrow 0 \mid \ldots \mid 9$$

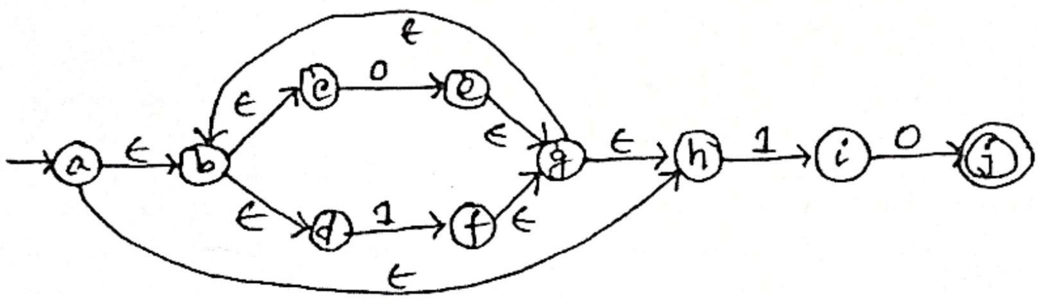
Whether the above grammar is ambiguous or not. Justify your answer using the derivation or parse tree for the input string "5*2+3-5". If it is ambiguous, convert it into unambiguous grammar using appropriate variables. Construct the syntax tree and left-most derivation for the input string "5*2+3-5".

b) Define a CFG for the language of 0 and 1 to generate the palindrome string. Explain the roles of the lexical analyzer and parser in a compiler with the appropriate block diagram. | 4 (2+2) | CO2 | Ap

c) Define all four basic types of grammar by emphasizing the production rules of the grammar with appropriate examples. | 2

3. a) Write the regular expression and draw the nondeterministic finite automata (NFA), deterministic finite automata (DFA) transition diagram, ε-NFA and CFG to define a identifier for C programming language (Σ={`a-z, A-Z, '_', 0-9`}) transition diagram to define the number of the C programming language. | 6 (1+1+1+1.5+1.5) | CO2 | Cr

b) Consider the following $\epsilon - NFA$ to define the all language (Σ={0,1}) end with one and zero. Convert it into the DFA using the ϵ-closure technique step by steps: | 4 (2+2) | CO2 | A



Or,

Differentiate between DFA and NFA. Describe the language (Σ={a,b}) in and $\epsilon - NFA$ for the regular expression: `a(a|b)*a` | 4 (1+3) | CO2

END

International Islamic University Chittagong
Department of Computer Science and Engineering
Program: B.Sc. in CSE: Semester: 5th
Mid-Term Examination, Autumn-2025

Course Code: CSE 3523 Course Title: Microprocessor, Microcontroller and Embedded System
Time: 1 hour 30 minutes Full Marks: 30

Answer the following Three (3) questions. Each question carries 10 marks.

		Questions	CLOs	Marks
1.	a)	Explain why the 8086 uses 20-bit address bus and what is its maximum addressable memory.	CO1	2
1.	b)	If Physical address is 3000FH, evalue the Segment and Offset address?	CO2	2
1.	c)	Explain the architectural evolution of Intel microprocessors from 8086 to Pentium, highlighting key improvements in architecture and performance.	CO1	6
2.	a)	"What are the four possible fields in an assembly language statement? Explain the purpose or function of each field.	CO2	4
		OR		
2.	a)	List and explain status flags of 8086.suppose,After an arithmetic operation, the 8086 Flags Register holds the value **0257H**.Write its binary equivalent and identify which status and control flags are set **(1)** and **reset (0)**.	CO2	4
2.	b)	I. MOV AL, [BX] ; II. MOV BL, [SI+3000H] III. MOV CL, [BX+SI] ; IV. MOV [DI], AL ; V. MOV [BX+DI], CL ; VI. MOV AL, [BX+DI+01H] For each instruction, determine (i) addressing mode, (ii) effective address, (iii) updated respective register and Memory value initial contents: **AX = 1234H**, **BX = 3001H**, **CX = 0002H**, **DX = 0000H**, **SI = 0003H**, **DI = 0004H**	CO2 SI DI	6
3.	a)	AX = OEFFH, BX = 125EH; Write an assembly language program to invert 5th and 8th bits, clear rightmost 2-bits and to set 13th and 15th bits. After that, obtain the final value of AX	CO2	5

Offset Address of DS	Data
3000H	22H
3001H	33H
3002H	44H
3003H	55H
3004H	66H
3005H	77H
3006H	88H
3007H	99H

<table>
<tr><td></td><td></td><td></td><td></td><td></td></tr>
<tr><td>3.</td><td>b)</td><td>What will be the final value of register and carry? Show the all shifting stages.
i) MOV AL, 9CH
SHL AL, 2H
ii) MOV AH, F7H
SAR AH, 3H</td><td>CO2</td><td>5</td></tr>
<tr><td></td><td></td><td>OR</td><td></td><td></td></tr>
<tr><td>3.</td><td>b)</td><td>I. For COUNT DB 10d, state the variable name, memory size, and the segment where it resides.
II. What is the purpose of the .MODEL directive in 8086 assembly programming
III. What does the .CODE and .DATA directive define in an assembly program?
IV. What is the function of the .STACK directive?
V. Differentiate between assembler directives and instructions

.MODEL SMALL
.STACK 100h
.DATA
COUNT DB 10d
LIMIT EQU 31
.CODE
MAIN PROC
MOV AX,@DATA
MOV DS,AX
INC COUNT
MAIN ENDP
END MAIN</td><td>CO2</td><td>5</td></tr>
</table>

International Islamic University Chittagong

Department of Computer Science and Engineering (CSE)

B.Sc. Engineering in CSE

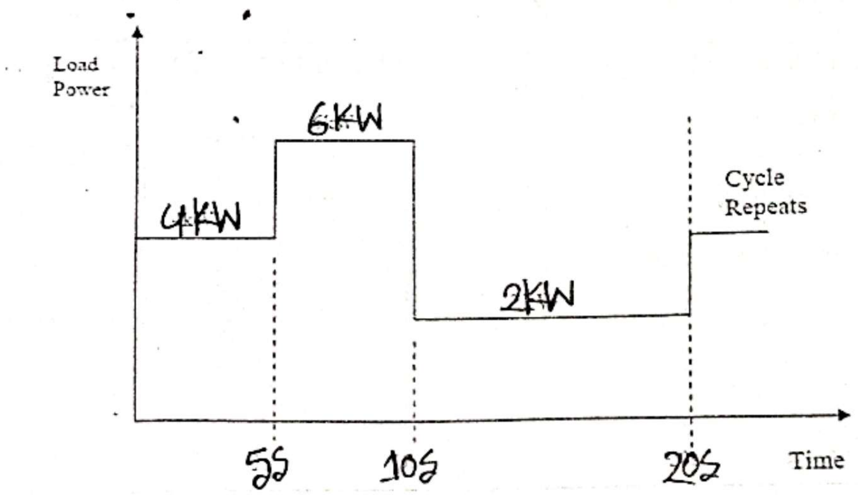
Midterm Examination, Autumn 2025

Course Code: EEE-2421 **Course Title:** Electrical Drives and Instrumentation

Time: 1.5 hours **Full Marks:** 30

Answer all the questions. The figures in the right-hand margin indicate full marks

Course Outcome (COs) and Bloom's Level (BL) are mentioned in additional Columns

1 (a)	**i.** Discuss the Linear DC machine as a motor with a proper diagram. **ii.** A coil that has 700 turns develops an average induced voltage of 50 V. What must be the change in the magnetic flux that occurs to produce such a voltage if the time interval for this change is 0.7 seconds?	CO1	U	3+2
	OR			
1 (a)	**i.** Discuss the Linear DC machine as a generator with a proper diagram. **ii.** Discuss the Hysteresis loss and the Eddy current loss.	CO1	U	3+2
1 (b)	The following load torque profile was measured for a constant speed load. Choose an appropriately sized motor to drive this Load. Assuming a service factor of 30%. You do not need to consider starting performance.  	CO3	Ap	5

2 (a)	Classify the types of transformers. Analyze the open circuit test and short circuit test of the transformer with the necessary diagram.	CO3	An	5
2 (b)	A 50-kVA, 4,400/220-V transformer has $R_1 = 3.45\ \Omega$, $R_2 = 0.009\ \Omega$. The values of reactance are $X_1 = 5.2\ \Omega$ and $X_2 = 0.015\ \Omega$. Calculate for the transformer: (i) equivalent resistance as referred to both primary and secondary (ii) equivalent reactance as referred to both primary and secondary (iii) equivalent impedance as referred to both primary and secondary (iv) total Cu loss, using equivalent resistances as referred to each side.	CO3	An	5
	OR			
2 (b)	**i.** Why is transformer rating in VA or KVA? **ii.** A 5 kVA 200/1000 V, 50 Hz, single-phase transformer gave the following test results: O.C. Test (L.V. Side): 200 V, 1.2 A, 90 W S.C. Test (H.V. Side): 50 V, 5A, 110 W Calculate the parameters of the equivalent circuit referred to the L.V. side.	CO3	An	1+4
3 (a)	i. Describe the operation of the DC machine that converts electrical power into mechanical power with a proper net sketch. ii. State Faraday's laws of electromagnetic induction, Fleming's right-hand rule and Fleming's left-hand rule.	CO1	R	3+2
3 (b)	i. Draw the Equivalent Circuit and Simplified Equivalent Circuit of a DC Motor with proper notations. ii. Discover the several magnetic field characteristics of a permanent magnet and an electromagnet with net sketches.	CO1	Ap	2+3

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International Islamic University Chittagong

Department of Computer Science and Engineering

B. Sc. in CSE, Midterm Examination, Autumn 2025

Course Code: **CSE-3529**

Course Title: **System Analysis and Design**

Total marks: **30**, Time: **1** hour **30** minutes

[Answer all the questions. Figures in the right-hand margin indicate full marks]

1. a) Discuss system analysis. Discuss the stages in building an improved system with diagram. 5 CO1

b) Write the sources of information. Discuss the quality/characteristics of information. 5 CO2

OR (of 1b)

Discuss the qualities of system analyst and perspectives of stakeholders on an information system.

2. a) What is SDLC? Briefly describe the different phases of SDLC with figure. 5 CO1

b) Briefly describe Waterfall model, Spiral model and Agile model. Which model is better? Why? 5 CO2

3. a) As we know to making feasibility study for any project is itself an expensive task and time-consuming matter. Now discuss the reason behind a project to perform feasibility study and cost benefit analysis. 5 CO1

b) A brief cash flow projection for an organization is as follows: 5 CO2

Description	Year 0	Year 1	Year 2	Year 3	Total
Total Benefits		45,000	50,000	57,000	152,000
Total Cost	100,000	10,000	12,000	16,000	138,000
Net Benefits (Total benefit- Total cost)	(100,000)	35,000	38,000	41,000	14,000
Cumulative Net Cash Flow	(100,000)	(65,000)	(27,000)	14,000	

Figure: Simple Cash Flow Projection

From the above conditions calculate the following:

(i) Return on Investment

(ii) Break-Even Point

OR (of 3b)

Describe the steps involved in performing an economic feasibility analysis of a project.